

Edge-AI Enabled Adaptive Traffic Signal Control with Vision-Based Emergency Vehicle Preemption

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Abstract—Urban traffic intersections experience rapidly changing traffic conditions that cannot always be handled efficiently by fixed-time signal plans. Recent developments in computer vision, edge computing, and artificial intelligence provide opportunities for real-time traffic perception and adaptive signal control. This paper proposes an Edge-AI enabled traffic signal framework that combines camera-based vehicle detection, lane-level traffic estimation, adaptive signal timing, and vision-based emergency vehicle preemption. Unlike a cloud-dependent architecture, the proposed framework performs the primary perception and decision-support operations close to the intersection, while cloud infrastructure is used optionally for long-term monitoring, analytics, and model maintenance. A vision module estimates vehicle counts, queue lengths, traffic density, and emergency vehicle presence from roadside camera streams. These measurements are provided to a signal-control module that dynamically selects signal phases while respecting minimum green, maximum green, yellow, clearance, and pedestrian-safety constraints. When an emergency vehicle is detected with sufficient confidence, a preemption request is generated and handled through the existing traffic-signal controller rather than directly bypassing controller safety logic. The proposed architecture also supports coordination across multiple intersections to form an emergency corridor. The paper defines a measurable evaluation framework based on vehicle delay, queue length, emergency vehicle delay, detection precision, inference latency, network usage, and signal recovery time. The framework provides a practical research direction for integrating computer vision and edge intelligence with established adaptive signal-control and emergency preemption technologies.

Index Terms—Edge AI, intelligent transportation systems, adaptive traffic signal control, computer vision, emergency vehicle detection, signal preemption, traffic monitoring, smart cities.

I. INTRODUCTION

Traffic congestion at urban intersections is influenced by variations in traffic demand, queue formation, road incidents, and changing vehicle arrival patterns. Conventional fixed-time traffic signals operate according to predefined timing plans and may therefore be less responsive when actual traffic conditions differ from the conditions used during timing-plan development. Adaptive traffic signal control addresses this limitation by using traffic measurements to modify signal timing according to current conditions [3].

Artificial intelligence has increasingly been investigated for traffic signal control. In particular, deep reinforcement learning (DRL) has been studied for learning traffic-control

policies from traffic-state information [1]. Research has also demonstrated the use of machine vision for estimating traffic-flow parameters and supporting adaptive signal timing [2]. These approaches indicate that camera-based perception can provide useful information for intelligent traffic management.

However, a practical traffic signal system must consider more than the accuracy of an AI model. Camera placement, illumination, weather, computational resources, communication delay, controller interfaces, pedestrian requirements, and failure handling can all affect system performance. Therefore, a complete architecture should separate traffic perception from safety-critical signal-controller logic.

Edge computing provides one possible architecture for this separation. Instead of transmitting every camera frame to a remote cloud server, an edge computer located near an intersection can perform vehicle detection and traffic-state estimation locally. Studies have demonstrated the feasibility of deploying deep-learning traffic detectors on resource-constrained edge hardware [6]. Local processing can reduce the quantity of raw video transmitted through a network and can allow the perception system to continue operating during temporary connectivity problems.

Emergency vehicle movement introduces an additional requirement. Emergency vehicle signal preemption is an established traffic management capability rather than a completely new function [4]. The research challenge is therefore not to claim that existing traffic systems cannot provide emergency preemption, but to investigate how reliable computer-vision detection and edge processing can be integrated with existing preemption mechanisms.

This paper proposes an integrated architecture combining four functions: (1) local traffic perception, (2) adaptive signal-control support, (3) emergency vehicle detection and preemption requests, and (4) coordination between multiple intersections. The objective is to establish a technically measurable framework that can be evaluated using real traffic video, simulation, or a hardware prototype.

The main contributions of this work are as follows:

- A modular Edge-AI architecture for intersection-level traffic perception and signal-control support.
- A lane-level traffic-state representation using vehicle count, queue length, density, and emergency vehicle presence.

- A safety-constrained adaptive signal-control strategy that distinguishes ordinary adaptive operation from emergency preemption.
- A multi-intersection coordination concept for creating an emergency vehicle corridor.
- A quantitative evaluation framework covering traffic efficiency, detection performance, latency, network usage, and recovery behavior.

II. RELATED WORK

A. Adaptive Traffic Signal Control

Adaptive signal control uses traffic measurements to modify signal timing according to changing traffic conditions. The Federal Highway Administration describes adaptive signal control as a technology in which traffic information is detected and used to adjust signal timing in real time [3]. Such systems depend strongly on the reliability of traffic detection.

Learning-based approaches have also been investigated extensively. Rasheed *et al.* reviewed deep reinforcement learning methods for traffic signal control and discussed their architectures, simulation environments, complexity, and research challenges [1]. Other research has investigated reinforcement learning methods for selecting signal phases and determining signal phase duration [7].

A limitation of many learning-based approaches is that simulation performance does not automatically demonstrate safe operation on physical traffic controllers. A deployable architecture therefore requires a clear interface between AI decisions and the established controller.

B. Computer Vision for Traffic Monitoring

Camera-based traffic monitoring provides a flexible source of traffic information. Vehicle detectors can estimate the number and type of vehicles approaching an intersection and can also provide information for queue estimation.

Gorodokin *et al.* investigated machine vision using YOLOv4 for estimating traffic-flow characteristics and adapting traffic light operation [2]. More recent edge-computing research has demonstrated real-time deployment of deep-learning vehicle detection on compact hardware. One study deployed YOLOv4 on an NVIDIA Jetson Nano and investigated traffic monitoring at an intersection [6].

These studies support the feasibility of local visual traffic processing. Nevertheless, practical performance can be affected by night-time conditions, shadows, rain, occlusion, camera angle, and vehicle appearance.

C. Emergency Vehicle Signal Preemption

Emergency vehicle signal preemption temporarily changes normal signal operation to provide right-of-way to an emergency vehicle. The FHWA Traffic Signal Timing Manual describes preemption as the transfer from normal signal operation to a special control mode and identifies multiple technologies for detecting preemption requests [4].

Emergency vehicle signal-control strategies have also been studied in intelligent transportation systems. An emergency-vehicle-centered signal-control method was proposed to reduce emergency vehicle delay while limiting disruption to conflicting traffic [8]. Another study proposed a real-time signal control strategy combining on-demand timing, signal preemption, and recovery after emergency vehicle passage [9].

These studies indicate that emergency vehicle prioritization must consider both the emergency vehicle and the traffic affected by the preemption operation.

D. Edge Computing for Traffic Intelligence

Edge computing moves computation closer to the data source. For traffic applications, an edge computer can receive video from a roadside camera, execute an object-detection model, and generate traffic-state information locally.

The advantage of this architecture is not that local processing automatically guarantees lower latency in every situation. Rather, local processing provides an architectural mechanism for reducing dependence on continuous cloud connectivity and reducing the transmission of raw video. Performance must still be measured for the selected hardware, model, camera, and communication configuration.

III. PROBLEM FORMULATION

Consider a signalized intersection with N incoming lanes. At time t , the traffic state of lane i is represented by

$$S_i(t) = [V_i(t), Q_i(t), D_i(t), E_i(t), W_i(t)], \quad (1)$$

where:

- $V_i(t)$ is the number of detected vehicles,
- $Q_i(t)$ is the estimated queue length,
- $D_i(t)$ is the estimated traffic density,
- $E_i(t)$ represents emergency vehicle presence, and
- $W_i(t)$ represents estimated waiting time.

The complete intersection state is

$$S(t) = [S_1(t), S_2(t), \dots, S_N(t)]. \quad (2)$$

The signal controller selects an action

$$A(t) = [P(t), G(t)], \quad (3)$$

where $P(t)$ represents the selected signal phase and $G(t)$ represents the corresponding green duration.

The objective of adaptive control is to reduce traffic delay and queue formation while maintaining safety constraints. A conceptual optimization function can be expressed as

$$J = \alpha D_{\text{avg}} + \beta Q_{\text{avg}} + \gamma W_{\text{avg}}, \quad (4)$$

where D_{avg} , Q_{avg} , and W_{avg} represent average delay, queue length, and waiting time, respectively. The coefficients α , β , and γ determine the relative importance of each term.

For emergency vehicle operation, the objective additionally includes emergency vehicle delay:

$$J_{EV} = J + \delta W_{EV}, \quad (5)$$

where W_{EV} is the waiting time of the emergency vehicle and δ represents its priority weight.

These equations define the research objective rather than claiming a particular experimentally measured performance.

IV. PROPOSED SYSTEM ARCHITECTURE

The proposed architecture consists of five major layers:

- 1) Camera sensing layer
- 2) Edge-AI perception layer
- 3) Traffic-state estimation layer
- 4) Signal decision layer
- 5) Optional cloud monitoring layer

The conceptual processing sequence is

$$\text{Camera} \rightarrow \text{Edge AI} \rightarrow \text{Traffic State} \rightarrow \text{Decision Module} \rightarrow \text{Signal Controller} \quad (6)$$

A. Camera Sensing Layer

Roadside cameras continuously capture traffic approaching the intersection. Cameras should be positioned so that the relevant lanes, stop lines, and approach areas remain visible.

The system does not require the cloud to process every video frame. Instead, frames can be processed by the local edge computer.

B. Edge-AI Perception Layer

The edge device performs object detection and tracking. A YOLO-family detector can be used to identify road users such as cars, buses, trucks, motorcycles, ambulances, fire engines, and police vehicles.

For each detected object, the system estimates:

- object class,
- bounding-box location,
- confidence score,
- lane assignment,
- movement direction, and
- tracking identity.

The edge device can therefore convert raw video into compact traffic measurements.

C. Traffic-State Estimation

Vehicle counts are calculated independently for each lane. Queue length can be approximated by identifying vehicles that remain within a predefined queue region.

A simple density measure is

$$D_i(t) = \frac{V_i(t)}{L_i}, \quad (7)$$

where L_i is the effective road-section length associated with lane i .

The estimated traffic state is updated periodically rather than necessarily for every camera frame.

D. Adaptive Signal Decision

The adaptive module uses the estimated traffic state to determine whether a signal phase should be extended, terminated, or selected.

A simple traffic-pressure measure can be defined as

$$P_i(t) = aV_i(t) + bQ_i(t) + cW_i(t), \quad (8)$$

where a , b , and c are configurable parameters.

The phase associated with a larger pressure value can receive a larger allocation subject to signal-safety constraints.

The algorithm must not bypass controller safety requirements. Minimum green, maximum green, yellow, all-red, pedestrian intervals, and other controller constraints remain enforced by the signal-control system.

E. Emergency Vehicle Detection

Emergency vehicles are detected using visual classification.

Detection confidence is represented by C_{EV} .

A preemption request is generated only when

$$C_{EV} \geq C_{\min} \quad (9)$$

and the detected vehicle satisfies additional temporal and spatial conditions.

A multi-frame confirmation mechanism can be used to reduce false detections. For example, an emergency vehicle may be accepted only when it is detected consistently over K consecutive frames.

F. Emergency Vehicle Preemption

When a valid emergency vehicle request is confirmed, the local system sends a preemption request to the traffic-signal controller. The controller remains responsible for executing the appropriate signal sequence.

This separation is important because emergency preemption is an established traffic-control mode. FHWA documentation specifies requirements concerning transition into and out of preemption and signal clearance intervals [4].

G. Multi-Intersection Coordination

For an emergency route containing several intersections, the system can estimate the emergency vehicle's direction of travel and expected arrival time.

For intersection k , an estimated arrival time can be represented as

$$T_k = \frac{d_k}{v_{EV}}, \quad (10)$$

where d_k is the estimated distance to intersection k and v_{EV} is the estimated emergency vehicle speed.

The coordination layer can then prepare downstream intersections while maintaining normal adaptive operation at intersections that are not currently being preempted. FHWA systems-engineering guidance also describes coordinated operation in which non-preempted signals continue adaptive operation while a preempted signal responds to an emergency request [5].

V. PROPOSED CONTROL ALGORITHM

The overall operation of the proposed framework is summarized in Algorithm 1.

Algorithm 1 Edge-AI Adaptive Traffic Signal Control

- 1: Capture traffic frame from roadside camera.
 - 2: Execute object detection on the edge device.
 - 3: Track detected vehicles across frames.
 - 4: Assign detected vehicles to lanes.
 - 5: Estimate vehicle count, density, queue length, and waiting time.
 - 6: Check for emergency vehicle detection.
 - 7: **if** emergency vehicle confidence $\geq C_{\min}$ **then**
 - 8: Confirm detection over multiple frames.
 - 9: Estimate emergency vehicle direction.
 - 10: Generate preemption request.
 - 11: Send request to traffic-signal controller.
 - 12: **else**
 - 13: Calculate traffic pressure for each approach.
 - 14: Select or extend the appropriate signal phase.
 - 15: Enforce minimum and maximum signal timing constraints.
 - 16: **end if**
 - 17: Record traffic and system measurements.
 - 18: Optionally transmit summarized data to the cloud.
 - 19: Repeat for the next processing interval.
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The algorithm separates emergency vehicle operation from ordinary adaptive traffic control. This distinction is necessary because preemption temporarily changes the normal signal-control mode [4].

VI. SAFETY AND RELIABILITY CONSIDERATIONS

A practical AI-based traffic system must consider failure cases. The following conditions are particularly important.

A. False Positive Detection

A false emergency vehicle detection could unnecessarily trigger preemption and disrupt traffic. Therefore, emergency detection should use confidence thresholds, temporal confirmation, and spatial validation.

B. False Negative Detection

Failure to detect an emergency vehicle may prevent the generation of a preemption request. Multiple camera views, improved training data, and additional sensing modalities can be investigated to reduce this risk.

C. Communication Failure

The system should not depend on continuous cloud connectivity for basic intersection perception. Local operation can allow the intersection to continue generating traffic measurements when the cloud connection is unavailable.

D. Edge Device Failure

If the AI device fails, the traffic controller should continue to operate using its established safe-control strategy. The AI system should therefore function as an intelligent input layer rather than becoming a single point of failure for the signal controller.

E. Controller Safety

AI decisions should be translated into requests that are accepted by the traffic-signal controller. Safety-critical signal sequencing should remain under the controller's validated logic.

VII. EXPERIMENTAL METHODOLOGY

The proposed framework can be evaluated using three experimental stages.

A. Stage 1: Computer Vision Evaluation

A labeled traffic-video dataset should be used to evaluate emergency vehicle and ordinary vehicle detection.

The following metrics should be measured:

$$Precision = \frac{TP}{TP + FP}, \quad (11)$$

$$Recall = \frac{TP}{TP + FN}, \quad (12)$$

and

$$F_1 = 2 \frac{Precision \times Recall}{Precision + Recall}. \quad (13)$$

Inference latency should also be measured on the selected edge hardware.

B. Stage 2: Traffic Simulation

A microscopic traffic simulator such as SUMO can be used to compare the proposed adaptive strategy with fixed-time signal control.

Traffic scenarios should include:

- low traffic volume,
- moderate traffic volume,
- peak traffic volume,
- uneven lane demand,
- emergency vehicle arrival, and
- multiple emergency vehicles.

C. Stage 3: Hardware-in-the-Loop Evaluation

A prototype can be constructed using an edge device, camera, and traffic-signal controller interface.

The evaluation should measure:

- average vehicle waiting time,
- average queue length,
- emergency vehicle waiting time,
- emergency vehicle travel time,
- detection precision and recall,
- AI inference latency,
- communication latency,

- network bandwidth usage,
- signal recovery time, and
- edge-device resource utilization.

No numerical performance claim should be made until these quantities are experimentally measured.

VIII. SYSTEM COMPARISON

Table I provides a conceptual comparison of the major approaches.

TABLE I
CONCEPTUAL COMPARISON OF TRAFFIC SIGNAL APPROACHES

Characteristic	Fixed	Adaptive	Proposed Edge-AI
Traffic sensing	Limited	Required	Camera/AI
Real-time adaptation	Low	Yes	Yes
Local AI processing	No	Optional	Yes
Emergency detection	External/optional	External/optional	Integrated
Emergency preemption	Possible	Possible	Integrated request
Cloud dependence	Low	Variable	Optional
Multi-intersection coordination	Possible	Possible	Supported
AI-based perception	No	Optional	Yes

The comparison is architectural rather than a claim that one approach universally outperforms another. Actual performance depends on traffic demand, detector quality, controller configuration, hardware, and deployment conditions.

IX. RESEARCH GAP

Existing research has independently investigated adaptive traffic signal control, deep reinforcement learning, machine vision, edge-based traffic monitoring, and emergency vehicle preemption [1], [2], [6], [8].

The remaining research challenge is the reliable integration of these components into a practical intersection-level architecture.

The following research questions are therefore proposed:

- 1) How accurately can emergency vehicles be detected using roadside cameras under different illumination and weather conditions?
- 2) What inference latency can be achieved on a resource-constrained edge device?
- 3) How should traffic density and queue length be converted into adaptive signal decisions?
- 4) How can emergency preemption requests be validated before being passed to a traffic-signal controller?
- 5) How should preemption requests be coordinated across multiple intersections?
- 6) How quickly can normal adaptive operation be restored after an emergency vehicle passes?
- 7) What are the traffic-flow and network-usage differences between local processing and cloud-based processing?

These questions provide measurable research objectives without assuming unsupported improvements in speed, cost, or accuracy.

X. EXPECTED CONTRIBUTION

The proposed system is intended to contribute an integrated architecture rather than a single AI detection model. Its principal contribution is the combination of local visual traffic perception, adaptive signal decision support, emergency vehicle detection, and multi-intersection coordination.

The architecture also provides a separation between AI perception and the safety-critical traffic-signal controller. This separation can support safer experimentation because the AI module generates traffic information and control requests while the controller continues to enforce its timing and safety constraints.

The proposed evaluation methodology further provides a basis for comparing different AI models, edge devices, traffic-control algorithms, and communication configurations using measurable performance indicators.

XI. LIMITATIONS

The proposed framework has several limitations.

First, camera-based perception can be affected by occlusion, illumination, weather, camera placement, and image quality. Second, edge devices have limited computational resources compared with cloud servers. Third, emergency vehicle classification based only on visual appearance may produce false detections when vehicle markings are unclear.

In addition, simulation results cannot fully reproduce the complexities of a physical traffic intersection. Controller interfaces, pedestrian movements, driver behavior, communication failures, and road geometry must be considered before field deployment.

Therefore, the proposed architecture should initially be evaluated in simulation and controlled prototype environments before any real-world deployment.

XII. CONCLUSION

This paper presented an Edge-AI enabled architecture for adaptive traffic signal control with vision-based emergency vehicle preemption. The proposed system processes traffic-camera information near the intersection and estimates vehicle count, traffic density, queue length, and emergency vehicle presence. These measurements support adaptive signal decisions while maintaining the distinction between ordinary adaptive control and emergency signal preemption.

Emergency vehicle preemption is treated as an established traffic management capability rather than as a new function. The proposed research direction focuses instead on reliable computer-vision detection, low-latency edge inference, safe communication with traffic-signal controllers, and coordination across multiple intersections.

Future work should implement the proposed architecture using an edge-computing platform and a traffic simulator, followed by controlled hardware evaluation. The resulting system should be assessed using detection accuracy, inference latency, vehicle delay, queue length, emergency vehicle delay, network usage, and signal recovery time. Such measurements

would provide an evidence-based assessment of the feasibility of Edge-AI traffic signal management.

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